

Bicontinuous Microarchitected Scaffolds Provide Topographic Cues That Govern Neuronal Behavior and Maturation

Prince D. Okoro, Kevin Dalsania, Shiril B. Iragavarapu, Benjamin Dela Cruz, Aihik Banerjee, Merve Basaranbilek, Martin F. Haase, Bahman Anvari, and Iman Noshadi*

3D tissue-engineered models hold great promise for recreating the intricate architecture and dynamic functions of neural tissues. However, replicating the nuanced structural cues of the brain in vitro remains challenging, as existing platforms often fail to capture the precise architectural motifs that regulate biological responses. Here, a bicontinuous interfacially jammed emulsion gel (bijel)-based fabrication strategy that combines solvent transfer-induced phase separation (STriPS), microfluidics, and bioprinting to develop a Bijel-Integrated PORous Engineered System (BIPORES) for neural tissue engineering is introduced. This multifaceted approach yields scaffolds featuring interconnected micropores and textured surfaces interspersed with a hyperbolic curvature, seamlessly integrated within macroscale fibrous networks. By leveraging STriPS of a ternary precursor mixture stabilized by amphiphilic nanoparticles, we synthesized poly(ethylene glycol) diacrylate (PEGDA) BIPORES support neural stem cell adhesion within 30 s without additional biological factors—a first for PEGDA scaffolds. Long-term cultures demonstrate extensive migration, robust proliferation, and differentiation into neuronal and astrocytic lineages, forming 3D networks with enhanced synaptic activity. Collagen encapsulation amplifies 3D cell growth, simulating native neuroanatomical compartmentalization. From a biomimicry standpoint, this multiscale fabrication strategy better approximates native neural tissue dynamics with significant implications for disease modeling, drug screening, and regenerative therapies.

1. Introduction

Neural tissue engineering seeks to recreate the complex extracellular matrix (ECM) environment of the brain that support the growth, differentiation, and functional integration of neuronal cells. This dynamic and highly organized microenvironment is central to the brain's intricate architecture, orchestrating cell behavior through precise structural and biochemical cues.^[1] Conventional 2D monolayer cultures, while indispensable for fundamental research, offer neither the structural features nor the ECM-mimetic milieu required for physiologically relevant neuronal maturation.^[2] Conversely, organoids capitalize on the self-organizing capabilities of pluripotent stem cells to form brain-like structures but suffer from high variability, limited vascularization, and incomplete functional maturation, confining their reliability for long-term neurological applications.^[3] Hydrogels do confer ECM-like properties yet often contract or degrade over time, curbing their ability to sustain stable and metabolically active neural constructs.^[4,5]

P. D. Okoro, K. Dalsania, S. B. Iragavarapu, B. Dela Cruz, A. Banerjee, M. Basaranbilek, B. Anvari, I. Noshadi
Department of Bioengineering
University of California
Riverside, CA 92521, USA
E-mail: inoshadi@ucr.edu

M. F. Haase
Van't Hoff Laboratory of Physical and Colloid Chemistry
Department of Chemistry
Debye Institute for Nanomaterials Science
Utrecht University
Utrecht 3584 CC, The Netherlands

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.202509452>

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Against this backdrop, the quest for stable and biomimetic 3D culture platforms remains at the forefront of neural tissue engineering, with porous scaffolds emerging as promising candidates. Established fabrication methods, including porogen and salt leaching,^[6,7] gas foaming,^[8] advanced bioprinting,^[9] and multilayer lithography,^[10] enable the fabrication of scaffolds with hierarchical porosity that optimizes bulk cell distribution and nutrient diffusion. Silk fibroin-based 3D constructs have shown particular promise owing to their interconnected porous structures, which facilitate neural stem cell proliferation, axonal growth, and ECM-like migration pathways.^[6,11] However, these approaches often emphasize macroporosity or employ nonintegrated hybrid strategies, which can hinder precise control over microstructural features.^[12] Moreover, reliance on ECM protein coatings for cell adhesion underscores a deficiency in intrinsic structural cues for biological recognition.

At the microscale, surface topography and geometry dictate neuronal cell behavior. Micropatterned grooves (10–20 μm wide) effectively guide axonal alignment and outgrowth,^[13] whereas surface roughness at both the nanoscale (≈ 30 nm)^[14] and microscale (1–5 μm)^[15] enhances integrin binding and axonal extension, fostering robust cell–substrate interactions. Complex geometries, notably surface curvatures, interconnected pore networks, and continuous channels influence cell polarity and migration. Progenitor cells have been shown to preferentially migrate toward concave surfaces owing to nucleus-mediated mechanosensing and focal adhesion remodeling,^[16,17] with concave microcurvature landscapes driving neuronal polarity and upregulation of differentiation markers.^[18] Likewise, bicontinuous microdomains facilitate 3D cell migration by providing interconnected pathways,^[19,20] whereas optimized pore sizes (5–50 μm) promote neural niche formation.^[21,22] A growing body of research shows that bicontinuous porous materials featuring continuous 3D channel networks and near-zero mean curvature consistently enhance diffusion of ions and small solutes.^[23–25] This geometry minimizes transport bottlenecks, making them particularly advantageous for neural tissue constructs where sustained nutrient delivery underpins long-term culture viability and functional maturation. Despite recent developments, scaling these structural motifs to tissue-level constructs without compromising the microarchitectural cues or structural integrity required for extended cultures remains a persistent challenge.

Here, we introduce a new bioactive material for neural tissue engineering, synthesized by integrating solvent transfer-induced phase separation (STriPS), microfluidics, and bioprinter-guided deposition to produce a 3D fibrous construct featuring bicontinuous microporous architectures. This builds upon the architectural paradigm of bicontinuous interfacially jammed emulsion gels (bijels), a soft matter system characterized by fully percolating, nonconstrictive pore networks with a uniform negative Gaussian curvature. Although well-established for applications in catalysis, separation, and energy storage,^[26–30] bijels are now emerging as viable platforms for biological applications.^[31,32] STriPS-based bijel fabrication especially enables the continuous production of microparticles, fibers, microropes, and films^[33–35] but has been limited to a thickness of ≈ 200 μm by diffusion barriers that intensify with material depth.^[36,37] Expanding on this approach, we developed a Bijel-Integrated

PORous Engineered System (BIPORES) that seamlessly integrates macroscale fibrous geometries with intricate microarchitectural details. BIPORES fibers were first synthesized by transforming a poly(ethylene glycol) diacrylate (PEGDA) precursor into interconnected porous frameworks with a negative Gaussian curvature, kinetically stabilized by colloidal silica nanoparticles. Using a custom microfluidic device paired with a bioprinter for spatiotemporal deposition, we produced 3D constructs featuring hierarchical interconnected pore networks that facilitate efficient nutrient diffusion, waste removal, and deep cell infiltration. These engineered architectures fostered neural stem cell adhesion, homing, robust proliferation, 3D axonal outgrowth, and synaptic activity, positioning BIPORES as a promising platform for advancing neural tissue engineering and regenerative therapies.

2. Results and Discussion

2.1. Continuous Fabrication of PEGDA-BIPORES Fibers

PEGDA-BIPORES fibers were fabricated using the STriPS technique, which transforms a homogenous ternary liquid solution into a bicontinuous interfacially jammed emulsion under controlled flow conditions.^[33,38,39] In this study, the precursor solution comprised PEGDA (M_n 250), ethanol, and water. Here, PEGDA acts as the oil phase due to its low miscibility with water, while ethanol serves as the solvent, facilitating the homogeneous mixing of the two immiscible phases. Details on the ternary phase diagram construction and identification of the critical point are provided in Figure S1 of the Supporting Information. Upon contact with a flowing aqueous NaCl phase, the rapid exchange of ethanol and water drives the mixture composition below the binodal curve, triggering spinodal decomposition and the formation of a bicontinuous network (Figure 1A). The stabilization of the transient bicontinuous structure relies on amphiphilic silica nanoparticles (SiNPs) functionalized via cetyltrimethylammonium bromide (CTAB) attachment. Initially hydrophilic, the SiNPs acquire amphiphilic characteristics upon CTAB functionalization, enabling their migration to the newly formed water–oil interfaces. Once localized, these interfacially jammed nanoparticles arrest droplet coalescence and preserve the bicontinuous architecture. In parallel, CTAB serves to disperse the particles uniformly within the ternary mixture, thus bolstering the overall interfacial stability (Figure 1B).

For continuous fiber production, a custom-built microfluidic device consisting of two concentrically aligned capillaries was employed (Figure 1C; Figure S2, Supporting Information). The innermost capillary delivers the ternary precursor solution, whereas the outer capillary introduces the NaCl phase coaxially (Figure S3, Supporting Information). As the precursor flow meets the NaCl stream, ethanol rapidly diffuses out and water diffuses in, driving the mixture into the spinodal decomposition region. This phase separation yields a bicontinuous matrix stabilized by the “jamming” of SiNPs at the interface. To maintain continuous fiber extrusion, the capillary walls were coated with a cationic polyelectrolyte, poly(diallyldimethylammonium chloride) (PDADMAC), to mitigate fouling and sustain laminar coflow (Figure S4, Supporting Information). Following extrusion, the resulting fibers underwent photopolymerization

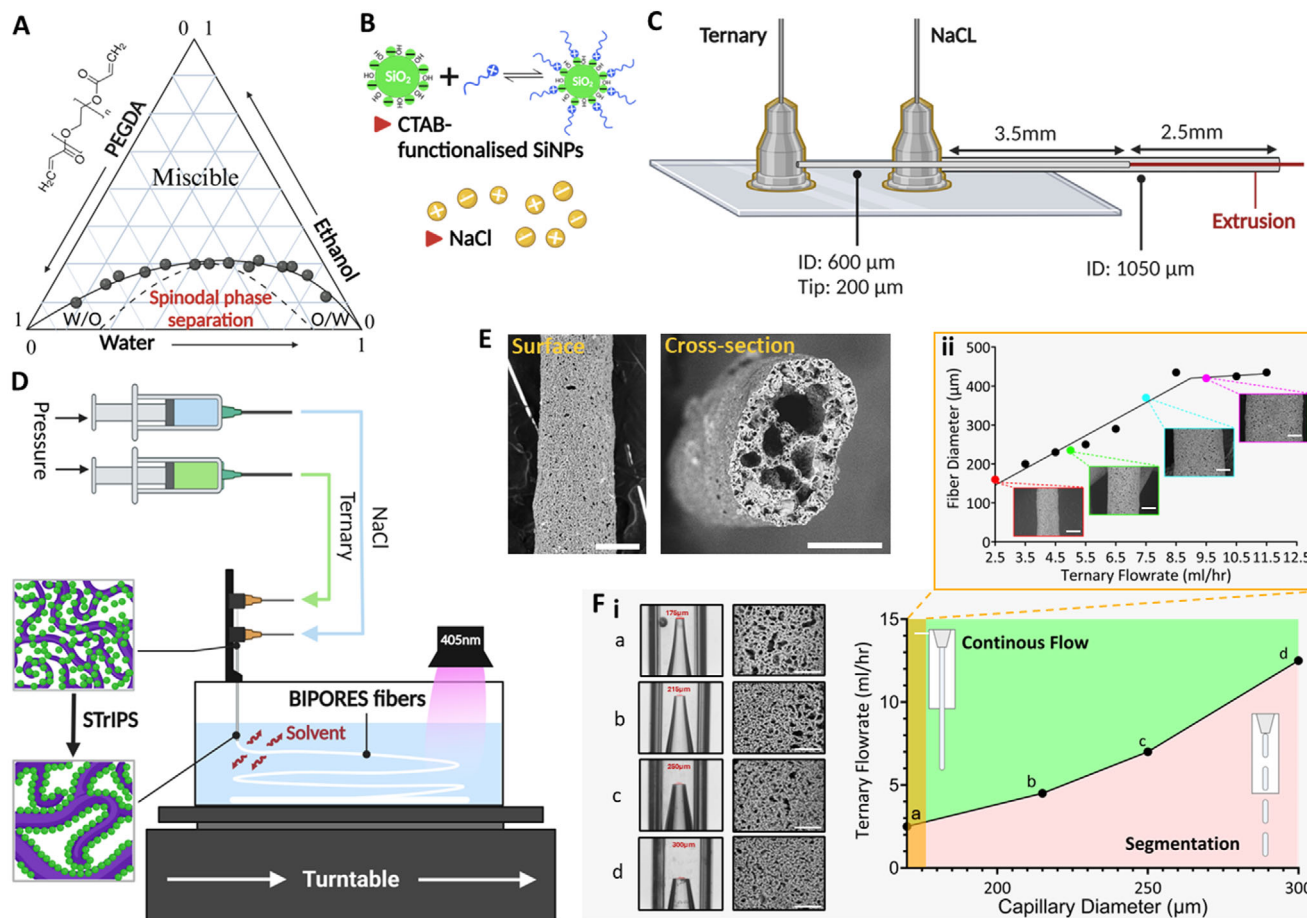


Figure 1. PEGDA-BIPORES fiber formation via StrIPS and flow-controlled morphology. A) Equilibrium phase diagram of the ternary solution consisting of PEGDA, ethanol, and water (40 vol% PEGDA, 30 vol% ethanol, 30 vol% water), highlighting the spinodal decomposition region crucial for bicontinuous phase formation. B) Schematic representation of CTAB-functionalized silica nanoparticles that stabilize the bicontinuous structure through interfacial jamming during phase separation. C) Fiber extrusion performed using concentric glass capillaries (inner diameter: 600 μm ; outer diameter: 1050 μm). The precursor mixture flows into a 100 mM NaCl aqueous stream, initiating StrIPS to produce BIPORES fibers. D) Solvent transfer drives spinodal decomposition, forming bicontinuous oil-water networks stabilized by interfacially jammed nanoparticles. Fibers are photopolymerized using 405 nm UV light with LAP as a photoinitiator and are collected on a rotating turntable for continuous production. E) Representative SEM images of the resulting BIPORES fiber surface (left) and cross-section (right). Scale bar: 100 μm . F) (i) Operational diagram of flow regime transitions illustrating the effect of ternary flow rate and capillary diameter on fiber morphology. Images show the resulting structural variations due to different operational conditions. Scale bar: 40 μm (ii) Inset demonstrating control of fiber diameter using an internal capillary diameter of 175 μm by modulating the ternary flow rate. Representative images further highlight the morphological changes under specific flow conditions. Scale bar: 100 μm .

through exposure to 405 nm visible light at an intensity of 30 mW cm^{-2} for 5 s. This step crosslinks the PEGDA polymer, converting the fluid-like bicontinuous system into a dimensionally stable, self-supporting scaffold that retains its bicontinuous architecture while enabling handling and cell culture manipulations (Figure 1D,E). Cross-sectional analysis of the fiber shows signs of a parabolic pore size profile (Figure 1E), characteristic of StrIPS-bijel systems. This radially-dependent morphology, characterized by smaller pores near the fiber surface and larger, more coarsened domains toward the interior, is attributed to differential rates of solvent exchange during the phase separation process.^[40]

A critical factor in fabricating PEGDA-BIPORES scaffolds is the continuous extrusion of single fibers. Beyond maintaining uninterrupted deposition, tuning the fiber diameter offers significant advantages in mimicking ECM structures, vascular tissue engineering, controlled drug release, and optimizing en-

ergy storage applications.^[41–44] The fiber diameter is controlled by adjusting the taper size of the microfluidic device's inner capillary; therefore, we selected the inner capillary taper diameter as a key control parameter. Additionally, the continuity, microstructure, and fiber diameter of PEGDA-BIPORES fibers are influenced by shear stress and the extent of solvent transfer during fabrication. Increased solvent transfer renders the fiber more viscoelastic and porous, heightening its susceptibility to breakage.^[45] Since modulating the solvent transfer rate can lead to undesirable BIPORES formation, we focused on controlling shear stress as the secondary control parameter. The shear stress is directly affected by the relative flow rates of the ternary mixture and NaCl solution. By varying the ternary flow rate while maintaining a constant flow rate in the continuous NaCl phase, we effectively regulate the shear stress exerted on the fibers.

Figure 1F(i) depicts the threshold for a continuous flow rate as a function of the inner capillary diameter in the PEGDA-BIPORES system and workflow described in Figure S5 of the Supporting Information. When the flow rate is too low for a given capillary diameter, segmentation occurs instead of continuous fiber formation (pink region, Figure 1F(i)). Increasing the flow rate above the critical threshold (green region) ensures the maintenance of a continuous fiber that can be extruded without disruption. Furthermore, the capillary dimensions coupled with the flow rate for continuous fiber extrusion appear to influence the final microstructure of the fibers. This effect is likely rooted in the kinetics of STriPS, where the final pore architecture is determined by the rate of solvent exchange before the structure is kinetically arrested by nanoparticle jamming at the liquid-liquid interface.^[46] We observed that a smaller diameter capillary operating at lower flow rates yielded fibers with a larger, coarser pore morphology (Figure 1F(i)). A plausible explanation for this result is the combination of rapid solvent exchange across the jet's high surface area-to-volume ratio and extended residence time, which likely allows sufficient time for domains to coarsen before kinetic arrest. Conversely, the finer pore architecture produced by larger capillaries at higher flow rates is consistent with a diffusion-limited solvent exchange.^[40] Here, the shortened residence time and solvent exchange likely induce a rapid quench, which may arrest the structure at an earlier, less coarsened stage.

Interestingly, at a constant capillary diameter of 175 μm , increasing the ternary flow rate from 2.5 mL h^{-1} to 12.5 mL h^{-1} proportionally increases the fiber diameter from ≈ 150 to 500 μm (Figure 1F(ii)), with no discernible change in microstructure. This degree of tunability over the fiber diameter and pore microstructure makes PEGDA-BIPORES fibers highly versatile materials for tissue engineering scaffolds, controlled drug delivery systems, catalytic membranes, and advanced filtration devices.

2.2. Microarchitecture Tuning via Compositional Variations

Bijel materials typically feature negative Gaussian curved surfaces formed via spinodal decomposition, producing extensive interfacial areas that are conducive to 3D cell infiltration and nutrient transport. A recent investigation of bijel-templated materials reported rapid tissue integration, largely driven by surface-mediated cues that enhance cell migration and nutrient diffusion.^[32] Here, we investigated how varying CTAB concentration (C_{CTAB}), SiNP wt. fractions (W_{NP}), and NaCl concentration (C_{NaCl}) modulate BIPORES surface features. Figure 2A maps the microstructural evolution as a function of CTAB concentration and SiNP loading. Bicontinuous morphologies where interpenetrating water/oil domains persist are confined to two distinct compositional windows: 40 mM CTAB with 10–25 wt% SiNP, and 50 mM CTAB with 15–25 wt% SiNP (red box). Within these regimes, the system exhibits predictable structural refinement with increasing particle content. For instance, at 40 mM CTAB, the mean pore diameter systematically decreases from $18.57 \pm 8.32 \mu\text{m}$ at 10 wt% SiNP to $7.01 \pm 4.37 \mu\text{m}$ at 25 wt% SiNP (Figure S6, Supporting Information), consistent with the inverse scaling of domain size with SiNP fraction commonly reported for STriPS-bijels.^[33,36] Outside this bicontinuity window, the CTAB–SiNP coupling becomes nonmonotonic, producing

separated voids, finger-like morphologies, or coarsened domains rather than bicontinuous channels. For instance, at 30 mM CTAB, the structure consists of small, separated voids that remain stable from 5 to 15 wt% SiNP (mean diameter $\approx 4 \mu\text{m}$) before undergoing sharp coarsening at 20 wt% ($11.90 \pm 7.32 \mu\text{m}$), suggesting insufficient surfactant to stabilize the additional interfacial area. This demonstrates that a delicate synergistic interplay is required to access predictable, interconnected porous architectures with the BIPORES platform.

From the structural mapping, the 40 mM CTAB series was identified as optimal C_{CTAB} to investigate the influence of surface architecture on cell response. Specifically, the CT40-10NP and CT40-15NP formulations were selected as they both exhibit well-defined bicontinuous networks yet possess distinct microarchitectures. This allows for a direct comparison of cell behavior on topographies differing in domain size. Confocal imaging (Figure 2B) and quantitative analysis (Figure 2C) confirm this distinction, with CT40-15NP presenting a finer, more uniform network (mean pore diameter: $11.58 \pm 5.21 \mu\text{m}$) compared to the larger, more heterogeneous pores of CT40-10NP ($18.57 \pm 8.32 \mu\text{m}$). While the 50 mM CTAB with 25 wt% SiNPs also produced bicontinuous structures, this formulation was excluded due to heightened cytotoxicity concerns associated with high surfactant concentrations,^[47,48] and dense silica packing that imparts brittleness that complicates handling.

In systems with cationic surfactants like CTAB, increasing ionic strength via NaCl is known to screen electrostatic repulsions, lowering the oil/water interfacial tension (γ) and often leading to structural coarsening.^[49] Interestingly, our results demonstrate clear domain refinement as NaCl concentration increases from 12.59 to 20.98 mM (Figure 2D). Given the lower hydrophobicity of PEGDA (M_n 250) compared to conventional oil phases in bijels,^[36,50] we effectively start with a lower baseline γ , making the system less sensitive to the thermodynamic destabilization from γ reduction. Here, the dominant effect of charge screening appears to be the kinetic acceleration of particle adsorption; this screening enhances CTAB packing on SiNPs, potentially rendering them more hydrophobic and promoting attachment.^[51] The accelerated rate could foster enhanced interfacial jamming, which arrests phase separation earlier, resulting in the observed finer pore morphology. Moreover, the larger 42 nm SiNPs used in this study possess a significantly higher intrinsic attachment energy ($\Delta E_{\text{attach}} \propto r^2$),^[52] creating a stability buffer that prevents interfacial collapse despite the modest reduction in γ . This combination allows the kinetic effect of accelerated jamming to dominate over the thermodynamic effect of γ reduction. This phenomenon is most apparent at 10 wt% silica, where the particle concentration is optimal for observing this kinetic tuning, whereas at 15 wt%, the particle coverage is relatively higher and the interface jams almost immediately, producing finer and more stable structures even at low C_{NaCl} . The proposed mechanism for this charge screening effect is schematized in Figure 2E.

In the continuous phase, NaCl acts as a “salting-out” agent, accelerating solvent exchange that drives the STriPS process. The dissolved ions reduce the thermodynamic activity of water, which intensifies the chemical potential gradient driving ethanol out and water into the precursor stream.^[53,54] This provides a significant thermodynamic quench that plunges the sys-

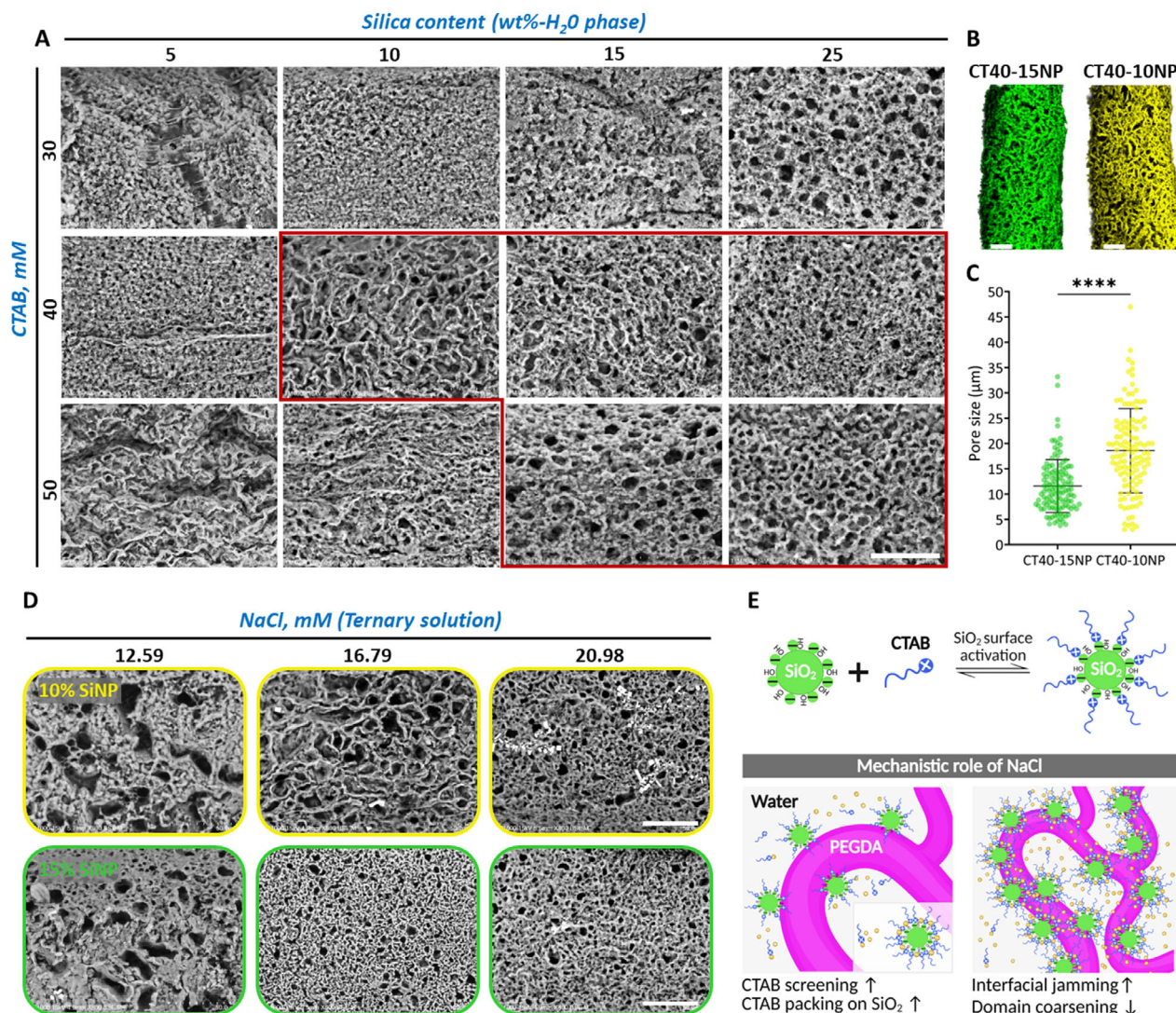


Figure 2. Microstructural tuning of BIPORES through CTAB, silica nanoparticle, and NaCl variations. A) SEM images illustrating the structural evolution of BIPORES formed at varying CTAB concentrations (30, 40, and 50 mM) and silica nanoparticle content (5, 10, 15, and 25 wt%). The red box highlights the region exhibiting the bicontinuous, interconnected morphology. Scale bar: 40 μm. B) 2D confocal projections provide an extended spatial perspective of the microstructures of CT40-10NP (green) and CT40-15NP (yellow). Scale bar: 40 μm. C) Quantitative pore size analysis of CT40-10NP and CT40-15NP showing larger and less uniform pores in CT40-10NP. Data are presented as mean ± SD to highlight variability. Statistical significance was determined using a two-tailed unpaired *t*-test ($***p \leq 0.001$). D) SEM images illustrating how NaCl concentration tunes domain size and network stability. Increased CTAB screening at higher NaCl concentration enables precise control over domain size and network stability, as demonstrated by the transition from larger domains to finer structures. Scale bar: 40 μm. E) Schema of NaCl-driven CTAB charge screening and its resulting effect on nanoparticle packing and pore morphology.

tem into the spinodal region more rapidly, shortening the coarsening time. The efficacy of this process and its dependence on SiNP content are clearly illustrated in Figure S7 of the Supporting Information. At a lower SiNP loading (10 wt%), where the system has less intrinsic stabilization, a strong thermodynamic quench from a high external C_{NaCl} is required to achieve a fine morphology before significant coarsening can occur. The images show a clear progression from coarse domains at 100 mM C_{NaCl} to a fine, uniform structure at 200 mM. Conversely, at a high loading (15 wt%), the system is already primed for interfacial jamming and more defined structures form even at a low

C_{NaCl} of 100 mM, as the system can rapidly arrest phase separation with only a modest thermodynamic push. At the highest C_{NaCl} for this loading, however, signs of coarsening emerge likely because the quench is overly aggressive and induces kinetic instabilities before the nanoparticle network can fully organize. This behavior highlights a synergy between the external thermodynamic drive from NaCl and the internal stabilization provided by SiNPs. These physicochemical parameters dictate BIPORES microstructures, providing a tunable framework for exploring the influence of surface features on cell interactions.

2.3. Single Cell–Material Interaction

Cell–material interactions are greatly influenced by surface features, chemical composition, and topographical cues.^[55] We employed real-time optical trapping to elucidate initial interactions between human iPSC-derived neural stem cells (i-HNSCs) with our materials. A laser optical trap allowed for the contact of individual cells with different test substrates under “tapping mode” (1 s contact) and “contact mode” (5, 15, and 30 s contact) (Figure 3A). Cell adhesion dynamics were then probed using a piezoactuated stage. Brightfield images (Figure 3B) revealed distinct cellular responses before and after contact with the test materials. i-HNSCs on PEGDA and ternary hydrogels showed no adhesion, as evidenced by the lack of cell anchorage even after 30 s of contact (Videos S2 and S4, Supporting Information). To determine if SiNPs alone could mediate cell adhesion, we investigated i-HNSC interactions with SiNP-coated PEGDA (Figure 3B). This was informed by studies demonstrating that SiNPs can promote cell attachment on otherwise bioinert surfaces through nanotopographical cues,^[56] enhanced protein adsorption,^[57] or direct physical interactions with the cell membrane.^[58] Despite this precedent, our real-time optical trapping experiments revealed SiNP-coated PEGDA provided insufficient cues for stable i-HNSC adhesion (Figure 3B). While a slight adhesive resistance was observed during material retraction (Video S6, Supporting Information), this transient interaction was too weak to withstand the dislodging force (F_{dis}) of the optical trap. These results indicate that SiNPs alone provide only a baseline for weak, nonspecific interactions and are insufficient to stably anchor i-HNSCs in this system.

In stark contrast, i-HNSCs on BIPORES fibers (CT40-15NP and CT40-10NP) exhibited strong adhesion as demonstrated by persistent interfacial engagement, firm anchorage, and cell deformation following the displacement of the test materials from the optical trap (Figure 3B; Figure S8 and Videos S8 and S10, Supporting Information). The minimum adhesion force (F_{mad}) for i-HNSC–BIPORES interaction was determined to be 1.92 ± 0.17 nN ($n = 3$). Given that PEGDA is a synthetic polymer devoid of intrinsic bioactive motifs, the observed i-HNSC adhesion on BIPORES fibers suggests a synergistic effect of SiNPs and surface topography. Specifically, the concave microcurvatures inherent to the bicontinuous architecture likely play a significant role, as similar features have been shown to promote cell attachment by guiding focal adhesion formation and minimizing cytoskeletal stress.^[59,60] In contrast to conventional PEGDA hydrogels, which often require functionalization with RGD peptides or ECM proteins (e.g., laminin, fibronectin),^[61,62] the structural features of BIPORES fibers appear to impart sufficient matrix-like cues for focal adhesion formation and cytoskeletal reorganization in neural stem cells.

To contextualize these findings, we performed adhesion assays on control substrates coated with laminin, the canonical ECM protein for neural cultures. As shown in Figure 3B and Video S12 (Supporting Information), laminin-coated PEGDA failed to support i-HNSC adhesion. This result is consistent with the well-documented protein-repellent or “antifouling” properties of PEGDA hydrogels, which resist nonspecific protein adsorption and subsequent cell attachment.^[63] Interestingly, i-HNSCs readily adhered to laminin-coated Ternary substrates (Figure S8 and

Video S14, Supporting Information). This indicates that the presence of SiNPs within the matrix facilitates the effective adsorption and presentation of laminin, a finding that aligns with studies showing silica nanoparticles enhance protein adsorption to form a bioactive “protein corona.”^[64] Furthermore, we evaluated i-HNSC adhesion to laminin-coated standard tissue culture plates (Laminin-TCPs) (Figure 3B; Video S15, Supporting Information). Adhesion was readily observed, confirming the attachment capability of i-HNSCs to Laminin-TCP. Such substrates are well documented to provide integrin-mediated cues that support rapid neural progenitor cell adhesion, sustained proliferation, and functional differentiation.^[65–67]

To investigate whether these initial interactions translate to mature cell–substrate attachment, we examined the cytoskeletal organization at 24 and 72 h (Figure 3C). i-HNSCs on PEGDA hydrogels displayed sparse actin networks and low cell coverage, whereas ternary groups promoted moderate cytoskeletal assembly, yet both remained substantially less pronounced than BIPORES. A similar outcome was observed with SiNP-PEGDA and laminin-PEGDA substrates (Figure S9, Supporting Information). Conversely, cells on BIPORES fibers formed extensive actin filaments with a marked increase in cell density, reflecting more stable adhesion and proliferative potential comparable to laminin-TCPs. These findings are consistent with the growing understanding that microtopographical features enhance focal adhesion formation and downstream mechanotransduction pathways.^[68] Correspondingly, Live/Dead assays demonstrated higher cell density on BIPORES relative to PEGDA or ternary surfaces (Figure S10, Supporting Information). The porous, interconnected channels of BIPORES presumably facilitate nutrient diffusion and waste removal, while the 3D curvature fosters stronger mechanical coupling with cell membranes. These morphological attributes enhance proliferation and guide differentiation in various progenitor cell lines, including neural stem cells, mesenchymal stem cells, and induced pluripotent stem cells.^[68–70]

2.4. Cytocompatibility Evaluation of PEGDA-BIPORES Fibers

A 21-day study was conducted to evaluate the cytocompatibility and biological response of i-HNSCs on BIPORES fibers. The CT40-15NP formulation was utilized owing to its uniform, interconnected micropores and processed via a three-day ethanol soak to remove residual CTAB. i-HNSCs were seeded directly onto BIPORES fibers at a density of $25\,000$ cells cm^{-2} without ECM protein coatings, and 2D polyornithine/laminin-coated culture substrates served as controls. Cells were monitored on Days 1, 7, 14, and 21 to capture early and late-stage responses to the engineered microenvironment. Live/dead staining (Figure 4B) showed predominantly live cells on both BIPORES and control surfaces from Day 1 through Day 7, with negligible cell death in both groups. Interestingly, cell migration from the well toward the fibers was observed after Day 7 in the experimental group, suggesting a preference for BIPORES fibers in the absence of ECM coating. By Day 14, there was an exponential increase in cell density, leading to the fibers being substantially covered by a proliferating cell mass (Figure 4B–D).

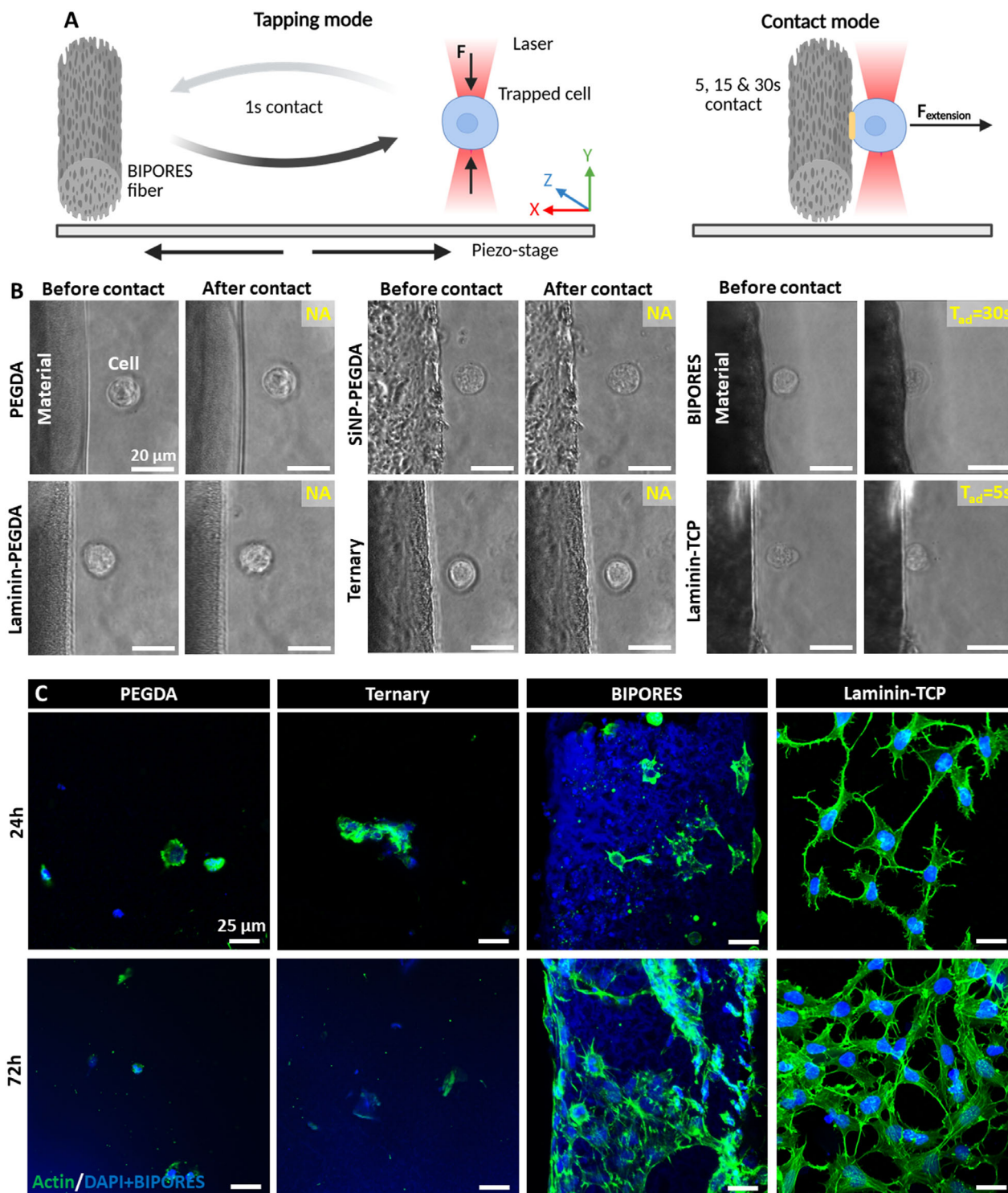


Figure 3. Assessment of single cell–material interaction. A) Experimental workflow for real-time evaluation of single-cell adhesion to biomaterial. A laser optical trap captures a single cell and brings it into contact with the material surfaces under tapping mode (1 s contact) or contact mode (5, 15, and 30 s). Adhesion forces are quantified with a piezostage for precise measurement. B) Phase-contrast images showing single-cell interactions with each material before contact and after optical-trap displacement following contact, with a laminin-coated tissue culture plate (Laminin-TCP) serving as a positive control. BIPORES (CT40-15NP) fibers exhibit enhanced adhesion, evidenced by visible cell deformation and strong interfacial contact, compared with PEGDA, SiNP-coated PEGDA, laminin-coated PEGDA, and Ternary surfaces (T_{ad} , time for adhesion; NA, no adhesion). Scale bar: 20 μ m. C) Fluorescence images illustrating cytoskeletal organization and nuclear distribution in adhered cells stained for actin (green) and DAPI (blue). At 24 and 72 h, BIPORES fiber and Laminin-TCP support pronounced cell adhesion, spreading, and proliferation, whereas PEGDA and ternary surfaces show limited cell interaction. Scale bar: 25 μ m.

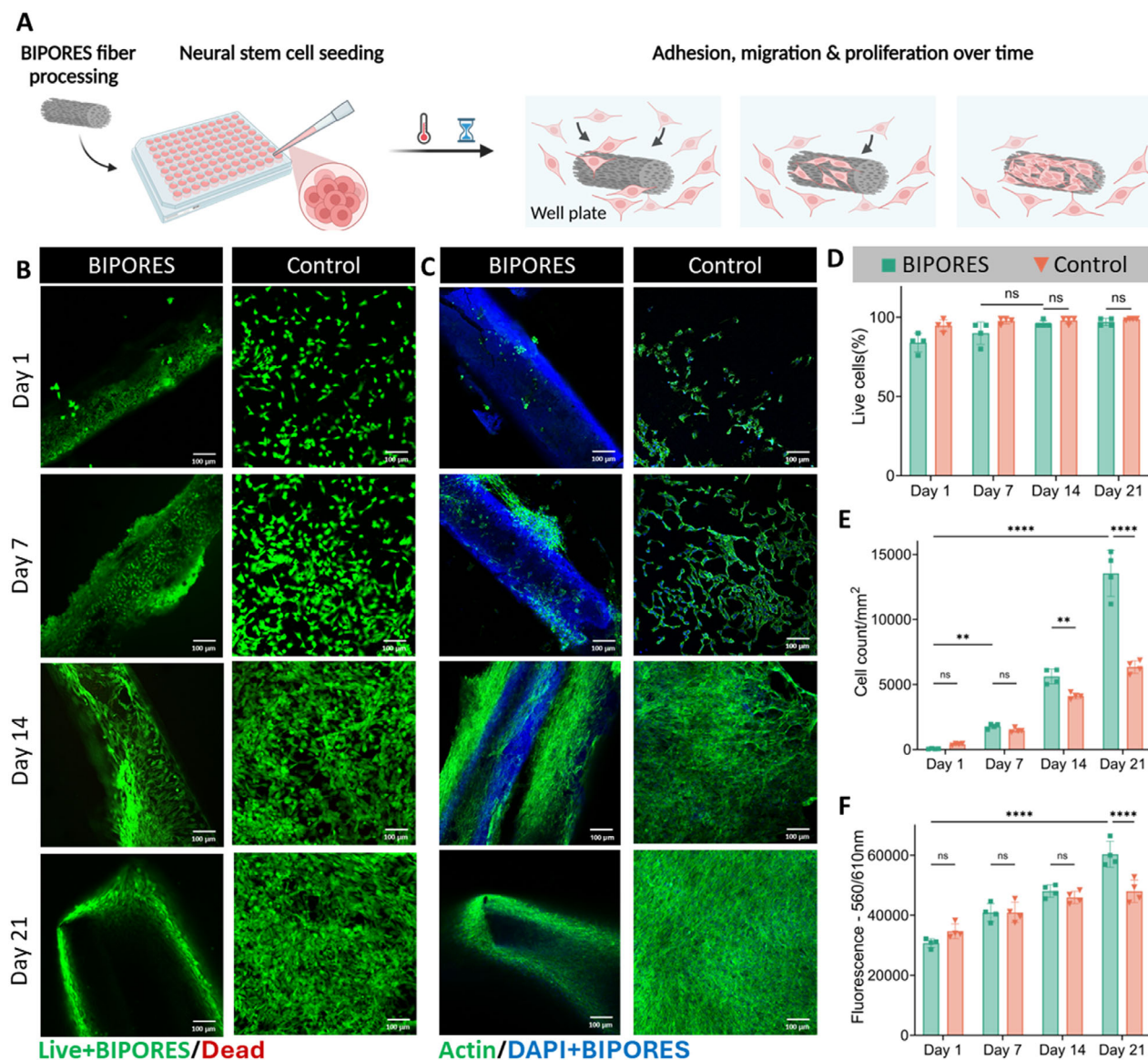


Figure 4. Cytocompatibility and morphological assessment of i-HNSCs on BIPORES fibers. A) i-HNSCs were seeded onto BIPORES fibers and ECM-coated 2D substrates to assess adhesion, migration, and proliferation over a 21-day period. The schema outlines the stages of preseeded fiber preparation, cell seeding, and postseeding analysis of cell behavior. B) Live/dead staining of i-HNSCs cultured on BIPORES fibers over 21 days. Live cells (green, calcein-AM) and dead cells (red, Ethidium Homodimer III) show high viability across all time points, comparable to the control group. C) Cytoskeletal organization and nuclear staining of i-HNSCs, with F-actin (green) and nuclei (blue), illustrating cell attachment and spreading on BIPORES fibers. Scale bar: 100 μ m. D) Quantification of live cell percentage demonstrates no significant difference between BIPORES and control groups over 21 days. E) Cell proliferation, assessed by DAPI-stained nuclei shows exponential growth over time, significantly higher on BIPORES fibers compared to controls from day 14. F) Metabolic activity, as indicated by fluorescence intensity increases steadily over the 21-day culture period, with significant differences observed on day 21. Data are presented as means $\pm$ SEM with statistical significance determined using a two-way ANOVA and Tukey's multiple comparisons test (** $p \leq 0.01$, **** $p \leq 0.0001$).

F-actin/4',6-diamidino-2-phenylindole (DAPI) staining provided additional insights into cell adhesion dynamics and proliferation (Figure 4C). On Day 1, characteristic rosette-like structures were observed on both BIPORES fibers and 2D substrates, indicating comparable initial adhesion. However, from Day 7 onward, cells on BIPORES fibers displayed enhanced proliferation, reflecting the role of interconnected pores and increased surface area in promoting cell-matrix interactions.

Porous textured microinterfaces mimic the interwoven meshwork of the native ECM, creating guiding paths for cell motility and colonization.^[20,71] Quantitative analysis of the cell count per unit area (Figure 4E) revealed a consistent increase throughout the study period. By Day 14, the cell density on BIPORES fibers rose sharply to 5619 ± 288 cells mm^{-2} , significantly exceeding the 4118 ± 115 cells mm^{-2} observed in controls ($p < 0.01$). The difference became more pronounced by Day 21, with BIPORES fibers

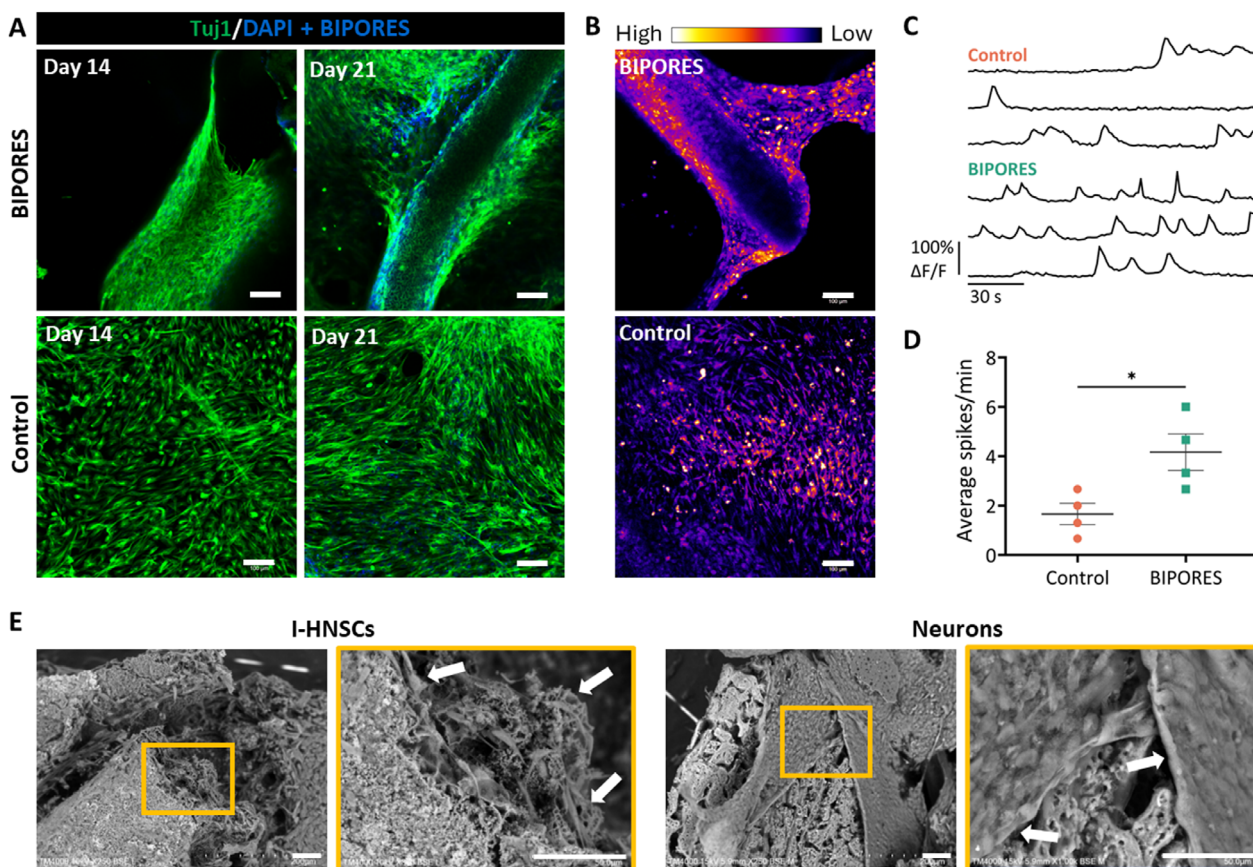


Figure 5. Neuronal differentiation, calcium activity, and infiltration of i-HNSCs on BIPORES fibers. A) Tuj1 and DAPI staining show neural differentiation on BIPORES fibers and ECM-coated substrates at days 14 and 21. Neurite outgrowth seems more pronounced on the fibers, with longer, dense projections, suggesting enhanced neuronal maturation compared to the control group. Scale bar: 100 μm . B) Heat maps of intracellular Ca^{2+} dynamics indicate higher baseline concentrations in BIPORES than in the control. Scale bar: 100 μm . C,D) Traces of Ca^{2+} fluctuations over time for both conditions show an increased frequency of calcium transients for cells on BIPORES. The spike rate is significantly higher for cells cultured on these substrates indicating enhanced neuronal activity. Data are presented as mean $\pm$ SEM from $n = 4$ biological replicates, with statistical significance determined by a two-tailed unpaired t -test ($*P \leq 0.05$). E) SEM images showing i-HNSCs and neurons infiltrating the interconnected microporous structure of BIPORES fibers after 14 days of culture. Confluent layers and deep infiltration into the fibers were visible, with cells depositing ECM within the porous network. Cell interaction with BIPORES appears to increase pore size and porosity, facilitating further cell infiltration and nutrient diffusion. Scale bar: 50 μm .

supporting 13547 ± 884 cells mm^{-2} , at least double the 6344 ± 239 cells mm^{-2} observed in the controls ($p < 0.0001$). These findings were corroborated by metabolic activity assays (Figure 4F), where the fluorescence intensity steadily increased over time, indicating enhanced cell viability on BIPORES fibers relative to 2D substrates. By Day 21, metabolic activity on BIPORES fibers reached 60313 ± 2154 RFU, significantly exceeding the 48031 ± 1887 RFU recorded for controls ($p < 0.0001$), reflecting the material's superior support for neural stem cell growth. The bicontinuous, interconnected microarchitecture of BIPORES scaffolds mirrors the native ECM, offering a more physiologically relevant environment than conventional 2D cultures.

2.5. Neuronal Differentiation, Maturation, and Integration within BIPORES Matrix

i-HNSCs were cultured on BIPORES fibers and polyornithine/laminin-coated 2D substrates in proliferation

media for five days before transitioning to differentiation media. Beta-3 tubulin (Tuj1) immunostaining at Day 14 confirmed the emergence of neuronal phenotypes in both culture conditions (Figure 5A). Notably, neurite extension appeared to be denser on BIPORES than on 2D controls. The 3D microarchitectures of BIPORES facilitate cytoskeletal rearrangements and growth cone extension, leading to the formation of long, branched neuronal processes. These effects are closely tied to enhanced integrin-mediated signaling, which is amplified in 3D environments owing to their biomimetic structural cues.^[72] By Day 21, neurite projections on the fibers became more interconnected, suggesting preliminary neural network formation that was less evident on the 2D control surfaces.

Functional assessments of neuronal differentiation were conducted via Ca^{2+} imaging on Day 28, capturing baseline intracellular calcium fluxes without external stimulation (Figure 5B–D). Cells grown on BIPORES exhibited elevated baseline Ca^{2+} levels and a higher frequency of spontaneous Ca^{2+} transients, indicating enhanced neuronal excitability and synaptic activity (Videos

S16 and S17, Supporting Information). The intricate, spiral interwoven channels within BIPORES scaffolds facilitate enhanced cell–cell and cell–matrix interactions, which are pivotal for neuronal differentiation and maturation. These structural features promote integrin clustering and the formation of focal adhesions, thereby modulating intracellular signaling pathways that govern calcium dynamics and synaptic protein maturation.^[73,74] Moreover, the hierarchical topography of BIPORES fibers provides both mechanical and spatial cues that drive cytoskeletal rearrangements, which are essential for the development of voltage-gated calcium channels and synapse formation. Indeed, recent studies have demonstrated that substrates with increased surface roughness and curvature significantly enhance synaptogenesis and neuronal excitability compared to flat surfaces.^[18,75]

Astroglial differentiation was evaluated via GFAP staining at Day 14, revealing characteristic stellate morphologies on both the BIPORES fibers and 2D substrates (Figure S11, Supporting Information). Although overall GFAP expression levels appeared comparable across conditions, astrocytes within the BIPORES adopted more aligned and elongated morphologies, consistent with reports that 3D topographies can direct glial cell orientation and regional specialization.^[76] By Day 28 (Figure S12, Supporting Information), brightfield imaging captured pronounced astrocytic alignment along the fibrous architecture, forming dense, organized layers compared to the 2D controls, which displayed random and scattered growth.

Scanning electron microscopy (SEM) images after 14 days of culture (Figure 5E; Figure S11, Supporting Information) revealed extensive cell coverage and infiltration, with i-HNSCs, neurons, and astrocytes encasing the BIPORES fibers. The interconnected microporous network facilitated deep cell infiltration into the scaffold and promoted ECM deposition, while localized cell-mediated remodeling appeared to enlarge pore sizes. Cross-sectional analyses revealed deep infiltration and the formation of multilayered cell aggregates within the scaffold, suggesting functional integration. These findings demonstrate that the bicontinuous domains of BIPORES fibers promote neuronal and astrocytic differentiation from i-HNSCs while simultaneously facilitating deep cell infiltration and scaffold remodeling. Enhanced neurite outgrowth, heightened baseline calcium signaling, and aligned astrocyte networks collectively indicate a more physiologically relevant environment compared to traditional 2D cultures.

2.6. Engineering Neurocompatible 3D Fibrous BIPORES Constructs

2.6.1. Fabrication and Processing of Millimeter-Sized BIPORES Constructs

The fabrication of 3D BIPORES constructs represents an evolution of STriPS-bijel systems, enabling the creation of millimeter-thick 3D scaffolds with tunable microarchitectural features. To achieve this, a custom-built microfluidic device was mounted on a bioprinter for spatial deposition of fibers in a layered meshwork configuration (Figure 6A; Figure S13, Supporting Information). Each extruded fiber preserved the characteristic bicontinuous subdomains, and stacking them resulted in

a macroscopic meshwork with interconnected porosity at multiple length scales. This hierarchical architecture was deliberately designed to foster bulk nutrient transport while retaining the characteristic spinodal microdomains necessary for cell anchorage and infiltration. CT40-10NP and CT40-15NP formulations were investigated to assess the potential influence of nanoparticle concentration on bioactivity. For CT40-10NP, a 200 mM NaCl concentration was selected for the continuous phase to achieve more uniform surface features, as previously shown (Figure S7, Supporting Information). Minimal morphological differences were observed between the two formulations, both exhibiting interconnected bicontinuous porosity integral to efficient mass transfer and robust cellular colonization (Figure 6B).

Long-term material stability is paramount for human neuronal cultures which require 60–100 days to achieve functional differentiation and synaptic network formation.^[77] A pH-dependent degradation study (Figure S14, Supporting Information) was performed to assess scaffold stability under both normal and pathological conditions. Over a two-month period, minimal weight loss (<0.3%) was observed across all groups, although a slight increase in degradation occurred under alkaline conditions (pH 11) compared to pH 3 and pH 7. This moderate alkali sensitivity likely stems from the hydrolysis of PEGDA acrylate ester linkages and SiNP dissolution at elevated pH, resulting in a marginally weaker structure.^[78,79] From a practical standpoint, stable performance at a physiological pH of 7 suggests suitability for extended neural cultures. Postfabrication processing was also optimized to eliminate residual CTAB, a vital stabilizer during the formation of bicontinuous domains, but a potential cytotoxic agent in biological applications. Dynamic ethanol washing under agitation at 37 °C (Figure S14, Supporting Information) notably reduced the amount of residual CTAB compared to a static soak at room temperature. For instance, in the CT40-15NP groups, dynamic processing reduced residual CTAB concentrations to 0.0701 ± 0.0014 mM, compared to 0.2147 ± 0.0013 mM using the static method. Similarly, in the CT40-10NP groups, dynamic treatment yielded residual CTAB levels of 0.0438 ± 0.0018 mM, which were markedly lower than the 0.2079 ± 0.0018 mM observed under static conditions. Interestingly, residual CTAB was consistently lower in the CT40-10NP scaffolds than in the CT40-15NP scaffolds with both methods. This is likely due to the lower nanoparticle content in CT40-10NP, which reduces the overall surface area available for surfactant adsorption, thereby facilitating its removal.

2.6.2. Neurocompatibility and 3D Neural Growth

After processing, i-HNSCs were seeded onto fibrous BIPORES constructs with dimensions of $1500 \mu\text{m} \times 2000 \mu\text{m} \times 1500 \mu\text{m}$. By Day 3, 2D confocal projections revealed extensive actin-stained cytoskeletal networks across both CT40-10NP and CT40-15NP scaffolds. Scaffolds with lower nanoparticle content (CT40-10NP) displayed higher cell density by Day 7, likely due to the reduced residual CTAB which enhanced cell viability and proliferation (Figure 6C,D; Figure S15, Supporting Information). To investigate how neural stem cells integrate microtopographical guidance with matrix-derived biochemical cues, we adapted

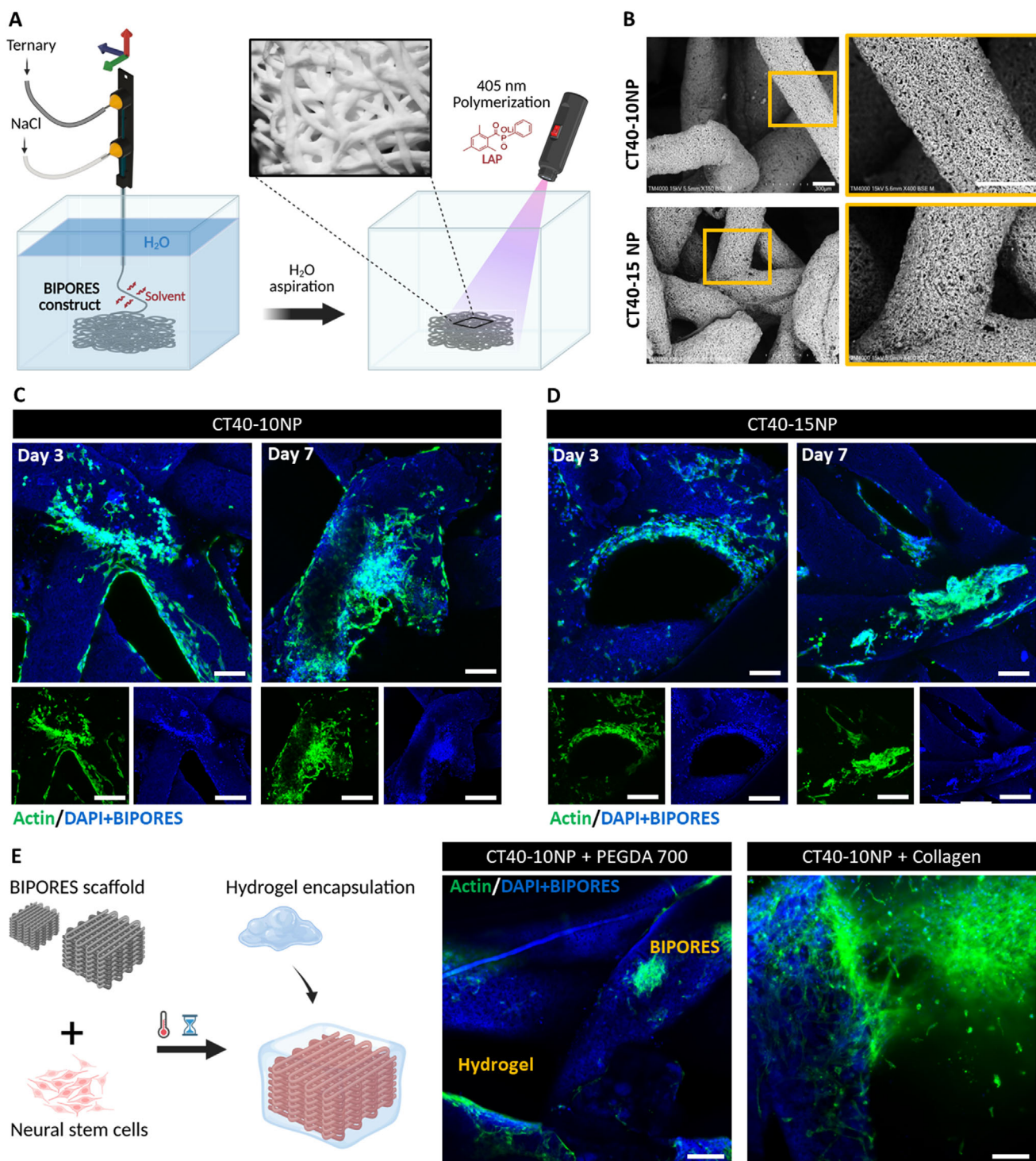


Figure 6. Bioprinter-aided fabrication and bioactivity of millimeter-sized BIPORES scaffolds. A) Spatial deposition of the precursor solution and NaCl via a microfluidic device into a nonsolvent bath, followed by photopolymerization under 405 nm UV light using LAP. Path-driven extrusion facilitates the fabrication of 3D constructs featuring a macroscale fibrous meshwork and interconnected porosity. B) Top-down SEM images of CT40-10NP and CT40-15NP scaffolds reveal a multi-level interconnected pore architecture. Scale bar: 100 μ m. C,D) Confocal images of i-HNSCs cultured on CT40-10NP (C) and CT40-15NP (D) scaffolds at Days 3 and 7. Actin staining (green) and DAPI-stained nuclei and scaffold (blue) illustrate robust cell attachment, spreading, and cytoskeletal organization, with CT40-10NP scaffolds exhibiting enhanced cell coverage compared to CT40-15NP scaffolds. Scale bar: 100 μ m; individual channels: 200 μ m. E) BIPORES scaffolds at Day 7 postencapsulation in collagen with PEGDA 700 serving as controls. 2D confocal projections highlight enhanced cell growth and extensions spanning both the collagen and BIPORES regions. Scale bar: 100 μ m.

BIPORES (CT40-10NP) constructs into a biphasic composite model as illustrated in Figure 6E. BIPORES were seeded with 5×10^5 i-HNSCs and encapsulated on day 5 within either a type I collagen hydrogel or a bioinert PEGDA (M_n 700) hydrogel, which served as a negative control. At Day 7 postencapsulation, confocal microscopy revealed marked morphological differences (Figure 6E). Cells on BIPORES constructs encapsulated in the collagen hydrogel extended numerous complex, 3D processes, forming interconnected networks that bridged the hierarchical BIPORES fibers and surrounding biochemically rich matrix. In contrast, cells within the PEGDA hydrogel remained largely confined to the scaffold surface with minimal growth and process extension (Figure 6E; Figure S15, Supporting Information). Such synergy reflects recent studies that combined porous silk scaffolds with collagen to enhance synaptogenesis and neural network formation.^[6,11] Here, the interplay between biochemical cues and microstructural guidance enhances the scaffold's potential to model key aspects of the complex microenvironment of the brain.

3. Conclusion

In this study, we engineered millimeter-scale PEGDA-BIPORES scaffolds using the STriPS technique, microfluidics, and bioprinting, seamlessly incorporating tortuous, bicontinuous microdomains within a robust 3D fibrous meshwork. By meticulously optimizing the flow rates, microfluidic device fabrication, and material composition, we fabricated microporous fibers with tunable surface architectures. Notably, neural stem cells adhered to BIPORES fibers within 30 s without the need for adhesion ligands or ECM coatings. This represents a novel achievement with PEGDA scaffolds, illustrating how engineered surface topographies can imbue bioactivity into an otherwise inert material.

In vitro studies revealed superior i-HNSC viability, extensive neurite outgrowth, and enhanced neuronal and astrocytic differentiation on BIPORES fibers compared to standard 2D substrates. The hyperbolic curvature and nonconstricting pore networks mirror the surface heterogeneity in neural microenvironments, providing structural cues that enhance cell–material interactions and functional maturation. Moreover, the fibrous BIPORES constructs showed great stability and neurocompatibility with collagen encapsulation augmenting 3D cell growth. This versatile platform offers a highly tunable architectural landscape across multiple length scales, making it an ideal template for the development of advanced in vitro neural tissue models and therapeutic scaffolds for neurological disorders.

4. Experimental Section

Materials: PEGDA (M_n 250), CTAB, lithium phenyl-2,4,6-trimethylbenzoylphosphine (LAP) (Cat. no. 900889), sodium chloride (NaCl), and 200 proof ethanol were purchased from Sigma-Aldrich (St. Louis, MO, USA). Colloidal silica nanoparticles (Ludox PW-50, 42 nm) were supplied by W.R. Grace (Columbia, MD, USA). PTFE tubing (Masterflex), 30 mL syringes (Global Medical Products), and generic 23 G Luer-lock dispensing needles facilitated solution delivery. PDADMAC (Sigma-Aldrich), hydrochloric acid (HCl) (Sigma-Aldrich), syringe pumps (KD Scientific), and Allevi 2 bioprinter (Allevi) were also utilized for material synthesis and processing.

Microfluidic Device Fabrication: A microfluidic housing was first fabricated using a Creality Ender 3 V2 3D printer with an Elegoo black polylactic acid filament. A custom-ordered borosilicate glass capillary (0.85 mm OD, 0.63 mm ID) was tapered using a microcapillary puller to achieve a fine tip; the final tapered diameter was measured and recorded. The microfluidic device was assembled by inserting this tapered inner capillary into the 3D printed housing, then sliding a square capillary (1.5 mm OD, 1.05 mm ID) over it for structural support and alignment. To complete the assembly, 23 gauge syringe tips were screwed onto the microfluidic housing. A two-part epoxy was applied at the critical junctions to seal the device, ensuring leak-free operation.

PEGDA-BIPORES Precursor Mixture and Fiber Extrusion: A ternary precursor mixture for PEGDA-BIPORES synthesis was prepared using PEGDA (M_n 250), ethanol, and an aqueous phase enriched with amphiphilic silica SiNPs at volume fractions $\phi_{\text{PEGDA}} = 0.4$, $\phi_{\text{ethanol}} = 0.3$, and $\phi_{\text{water}} = 0.3$ in a 10 mL solution. The aqueous phase (3 mL total volume) was formulated with precise compositions as follows: Ludox PW-50 SiNPs (50 wt% stock suspension) were first pH-adjusted from ≈ 10 to 3 using incremental 12 M HCl additions. To achieve the target 15 wt% SiNP loading in the final aqueous phase, 0.91 mL of this pH-adjusted SiNP suspension was combined with 1.4 mL of LAP solution (2.5 wt% in 16.79 mM NaCl). The remaining volume (0.69 mL) was completed with 16.79 mM NaCl solution to reach the total 3 mL aqueous phase volume. This aqueous mixture was thoroughly vortexed to ensure uniform SiNP dispersion. The ternary precursor was achieved by first combining 3 mL of this aqueous phase with 3 mL of ethanol containing 40 mM CTAB, followed by addition of 4 mL PEGDA (M_n 250). Vigorous vortexing produced a clear, homogeneous solution with well-dispersed nanoparticles, ensuring consistent interfacial stabilization during subsequent phase separation.

Prior to extrusion, a microfluidic device was coated with 1 wt% PDADMAC in 0.5 M NaCl for 2 h to prevent adhesion of the precursor mixture and then rinsed with DI water. Syringes containing the precursor mixture and a continuous phase of 16.79 mM NaCl were connected to the microfluidic device via PTFE tubing and mounted on programmable syringe pumps. The precursor mixture and continuous phase were pumped at 5 mL h^{-1} and 0.7 mL min^{-1} , respectively. During fiber extrusion, this specific differential flow induces a rapid exchange of ethanol with water, triggering solvent-mediated phase separation and the formation of bicontinuous domains stabilized by interfacially jammed nanoparticles. The resulting fibers were collected in a water-filled crystallizer on a rotating turntable to ensure alignment. Immediately after extrusion, the transient bicontinuous structure was photopolymerized by exposing the fibers to 405 nm UV light at 30 mW cm^{-2} for 5 s, crosslinking the PEGDA matrix and locking in the interconnected microporous architecture of the BIPORES fibers.

Taper Diameter and Flow Rate Calibration for Continuous Fiber Extrusion: Prior to testing, four microfluidic devices with taper diameters of 175, 215, 250, and 300 μm were prepared as previously described. Each device was initially operated with a ternary precursor flow rate of 2.5 mL h^{-1} and a static 100 mM NaCl flow rate of 0.7 mL min^{-1} . While keeping the continuous phase constant, the ternary flow rate was incrementally adjusted by 0.5 mL h^{-1} until a continuous flow was established. The critical flow rate at which continuous flow emerged was recorded. This cutoff was validated by temporarily reducing the flow rate to observe flow discontinuity and subsequently increasing it to confirm continuous flow. After three validation cycles, these optimized flow conditions were employed for BIPORES fiber synthesis. This procedure was repeated for each capillary diameter to evaluate the interplay between the taper size and flow rate for achieving continuous fiber extrusion.

Flow Rate Impact on Fiber Diameter: To examine the effect of flow rate on fiber diameter, experiments were conducted using a microfluidic device with a constant taper diameter of 175 μm . Starting with a ternary flow rate of 2.5 mL h^{-1} and a static 100 mM NaCl flow rate of 0.7 mL min^{-1} , BIPORES fibers were synthesized following the established protocol. Subsequently, the ternary flow rate was systematically increased in 1 mL h^{-1} increments across multiple additional runs, with a final synthesis at 5 mL h^{-1} . This series of experiments allowed for a systematic assessment of how variations in the ternary flow rate affect the resulting fiber diameter,

facilitating the optimization of fabrication parameters to achieve the desired scaffold features.

Precursor Mixture for CTAB-SiNP Study: To map morphological changes as a function of surfactant and nanoparticle concentration, a series of precursor mixtures was prepared by systematically varying the C_{CTAB} in the ethanol phase and the SiNP wt. fraction (W_{SiNP}) in the aqueous phase. Overall volume fractions were maintained constant at $\phi_{\text{PEGDA}} = 0.4$, $\phi_{\text{ethanol}} = 0.3$, and $\phi_{\text{water}} = 0.3$. The SiNP loading (5, 10, 15, and 25 wt%) refers to the wt. fraction of silica nanoparticles relative to the aqueous phase. To prepare a 3 mL aqueous phase with a specific W_{SiNP} , the required mass of SiNPs was calculated, and the corresponding volume of pH-adjusted Ludox PW-50 stock solution (50 wt%) was added to a constant 1.4 mL of LAP solution (2.5 wt% in 16.79 mM NaCl). The volume of the aqueous phase was then adjusted to 3 mL by adding 16.79 mM NaCl solution. This ensures that while the SiNP wt. fraction varies, the total aqueous phase volume and the absolute amount of LAP remain constant across all formulations. For each point in the C_{CTAB} versus W_{SiNP} phase map, the final 10 mL precursor mixture was prepared by combining 4 mL of PEGDA, 3 mL of the appropriate CTAB-ethanol solution, and 3 mL of the corresponding SiNP-aqueous phase, followed by vigorous vortexing.

Material Characterization: To evaluate the microstructural and topographical features of PEGDA-BIOPORES, SEM was performed using a Hitachi TM4000PlusE II microscope. Dried scaffolds were initially examined at low magnification to establish orientation and then imaged at higher magnifications for detailed surface morphology. Imaging was conducted in slow-scan mode at an accelerating voltage of 15 kV with backscatter electron detection, which enhanced contrast and revealed microstructural details. Confocal microscopy was performed using a Zeiss LSM 880 inverted confocal microscope. Z-stack images were acquired across the entire fiber surface, and 2D projections along with pore size measurements were obtained using ImageJ software.

For cell-laden fibers, a sequential dehydration and drying protocol was performed prior to SEM imaging. Samples were gradually dehydrated through a graded ethanol series: 50% for 5 min, 70% for 5 min, 80% for 10 min, 95% for 10 min, and 100% for 10 min, with the final step repeated twice to ensure complete water removal. Following ethanol dehydration, samples were treated with graded hexamethyldisilazane (HMDS, Sigma-Aldrich): immersion in a 3:1 ethanol:HMDS mixture for 15 min, then 1:1 for 15 min, followed by 1:3 for 15 min, and finally three successive immersions in 100% HMDS for 15 min each. After air-drying overnight, the cell-laden scaffolds were imaged via SEM using the same process as for the acellular samples.

Surface Pore Size Quantification: Surface pore size measurements were performed using ImageJ software on high-contrast SEM images. Images were calibrated using scale bars, converted to 8-bit grayscale format, and contrast enhanced with normalization to optimize pore-matrix distinction. Binary segmentation was achieved through manual thresholding with “dark background” selected to isolate surface pores from the surrounding polymer matrix. Particle analysis was performed with size constraints set to exclude imaging artifacts while maintaining circularity parameters at 0.00–1.00 to capture the full range of pore morphologies. For each formulation, measurements were obtained for ≥ 120 pores and values were exported statistical analysis. The mean pore diameter $\pm$ standard deviation was calculated to quantitatively characterize size distribution.

Degradation Test: The degradation profile of PEGDA-BIOPORES scaffolds was investigated under three pH conditions: 3, 7, and 11. Phosphate-buffered saline (PBS) solutions at these pH levels were prepared by adjusting with HCl or NaOH. For each pH condition, four samples (≈ 70 mg each) were weighed (initial dry weight, W_0), placed in centrifuge tubes, frozen at -80°C for 24 h, and subsequently freeze-dried for 24 h. After freeze-drying, each sample was immersed in their corresponding pH-adjusted PBS. At designated time points (Days 1, 7, 14, 28, and 60), PBS was removed, samples were refrozen and freeze-dried for 24 h, and their dry weights (W_t) were recorded. The percentage of weight loss, which represents degradation, was calculated using the following equation

$$\text{Weight loss (\%)} = \frac{W_0 - W_t}{W_0} \times 100 \quad (1)$$

Real-Time Assessment of Single Cell–Biomaterial Interactions: To evaluate single-cell adhesion and interactions with biomaterials, a custom laser tweezer system integrated into an inverted microscope (Nikon Ti-Eclipse) was employed as previously described.^[80] Three test materials were investigated: PEGDA-BIOPORES fibers synthesized via STriPS, a crosslinked ternary solution without STriPS, and PEGDA hydrogels. The ternary precursor and PEGDA formulations were prepared as previously described, with PEGDA hydrogels comprising 40% (v/v) PEGDA, 30% (v/v) ethanol, and 0.25% (w/v) LAP as the photoinitiator. Test substrates for the ternary solution and PEGDA hydrogels were fabricated on glass coverslips by UV crosslinking with 405 nm light for 5 s. SiNP-coated samples were prepared via drop-casting and evaporative deposition at 50°C overnight to ensure complete solvent removal. For relevant samples, laminin (Cell Applications) coating involved incubation for 2 h at 37°C , followed by a gentle PBS wash before testing. All materials were adhered to poly-D-lysine-coated dishes (MatTek) using a medical-grade adhesive (Factor II, Inc.) and air-dried for 24 h before experimentation. The i-HNSC growth medium was carefully added to the plates to avoid substrate displacement, and 1×10^5 cells were introduced. The cells were allowed to settle evenly across the dish prior to optical manipulation.

The laser tweezer system utilized a 1064 nm continuous-wave laser beam directed through a high numerical aperture objective lens to generate optical traps at the microscope stage. The tightly focused laser beam produced a high-intensity gradient force that immobilized individual i-HNSCs, allowing precise 3D manipulation and real-time observation of cell–substrate interactions. For all adhesion tests, a piezoelectric translation stage (Physik Instrumente) enabled the precise movement of the substrates toward and away from the optically trapped cells at a constant velocity of $10 \mu\text{m s}^{-1}$. Cell adhesion dynamics were analyzed under two configurations. In tapping mode, cells were briefly brought into contact with the substrate for 1 s to evaluate initial adhesion. With the exception of the Laminin-TCP, there was no cell adhesion in tapping mode for the test substrates (Videos S1, S3, S5, S7, S9, S11, Supporting Information). In contact mode, cells remained in sustained contact with the surface for durations of 5, 15, and 30 s, facilitating assessment of substrate-specific affinities through observations of cell anchoring and morphological changes.

The minimum adhesion force (F_{mad}) was quantified between i-HNSCs and BIOPORES. A trapped cell detaches only when the adhesive force exceeds the displacement force (F_{dis}), which is given by $F_{\text{dis}} = k r$, where k is the trap stiffness and r is the cell radius. The trap stiffness, calibrated by applying known viscous drag forces to a trapped microparticle, was $320 \text{ pN } \mu\text{m}^{-1}$. Using this stiffness and the measured cell radius, F_{mad} was calculated for the i-HNSC–BIOPORE interaction.

i-HNSC Maintenance: i-HNSCs were obtained from Cell Applications, Inc., and cells were used between passages 4 and 6 for all experiments. Following rapid thawing in a 37°C water bath, cells were diluted with pre-warmed i-HNSC Growth Medium, centrifuged at $250 \times g$ for 5 min and resuspended in 3 mL of i-HNSC Growth Medium (Cell Applications) supplemented with $10 \mu\text{M}$ ROCK inhibitor (Y-27632 Dihydrochloride, STEMCELL Technologies). Before cell seeding, six-well plates were sequentially coated with poly-L-ornithine for 1 h at 37°C , aspirated, then coated with laminin for 2 h at 37°C . After washing with sterile PBS, i-HNSCs were seeded at a density of 1×10^6 cells per well with a supplemented growth medium. The medium was renewed after 24 h to remove the ROCK inhibitor, and cultures were maintained at 37°C in a 5% CO_2 humidified incubator with the medium changed every third day until $\approx 90\%$ confluency. For routine passaging, cells were dissociated using StemPro Accutase (Gibco) at 37°C , harvested, centrifuged, and resuspended in i-HNSC Growth Medium with $10 \mu\text{M}$ ROCK inhibitor. Cells were reseeded at a density of 1×10^6 cells per well on freshly coated six-well plates, following the same coating and seeding procedures. This standardized protocol ensures consistent maintenance of i-HNSCs in a healthy, undifferentiated state, ready for downstream differentiation and seeding onto BIOPORES fibers.

i-HNSC Culture on PEGDA-BIOPORES Fibers: For cytocompatibility and differentiation studies with BIOPORES fiber, CT40-15NP BIOPORES were selected owing to their uniform microarchitecture and superior substrate interaction. Before cell seeding, the fibers were soaked in 200 proof ethanol

for three days at room temperature to remove residual CTAB, followed by serial PBS washes to eliminate residual ethanol. The fibers were then incubated in i-HNSC Growth Medium to equilibrate them for cell culture. Preliminary experiments identified a seeding density of 2.5×10^4 cells per well in a 48-well plate to achieve exponential cell growth over a 21 day period. For cell seeding, i-HNSCs were gently pipetted onto the pretreated fibers at the specified density and allowed to adhere for 20 min at 37 °C in a 5% CO₂ incubator. Subsequently, i-HNSC Growth Medium supplemented with 10 μM ROCK inhibitor was gently added. Media changes were performed every three days, with cytocompatibility and differentiation assessments conducted at designated time points. As a control, i-HNSCs were cultured in well plates coated with poly-L-ornithine and laminin. For differentiation studies, they were maintained in i-HNSC Growth Medium until Day 5, after which neural or astrocyte differentiation media were introduced for lineage-specific differentiation.

Cytocompatibility Assessments: The cytocompatibility of BIPORES fibers was evaluated using Live/Dead assays, actin/DAPI staining, and PrestoBlue metabolic activity measurements. The LIVE/DEAD Viability/Cytotoxicity Kit (Invitrogen) employs calcein-AM and ethidium homodimer-1 to distinguish live cells from dead cells based on intracellular esterase activity and plasma membrane integrity, respectively. A working solution was prepared by adding 20 μL of 2 mM EthD-1 and 5 μL of 4 mM calcein-AM to 10 mL of sterile, tissue culture-grade DPBS, yielding final concentrations of 2 μM calcein and 4 μM EthD-1. On days 1, 7, 14, and 21, growth medium was aspirated from i-HNSC cultures on BIPORES fibers, followed by a brief DPBS wash. The samples were then incubated in the Live/Dead working solution for 30 min at room temperature in the dark. Fluorescence imaging was performed using a Zeiss LSM 880 inverted confocal microscope. ImageJ software was used for quantitative analysis of the live cell percentage across four biological replicates.

To assess the cytoskeletal architecture and nuclear morphology of i-HNSCs cultured on BIPORES fibers, cells were stained with fluorescently conjugated phalloidin to visualize filamentous actin and DAPI for nuclear labeling. Samples were initially washed twice with PBS (pH 7.4, Gibco) and fixed with 4% (v/v) paraformaldehyde (PFA, Thermo Scientific) for 15 min at room temperature. Following fixation, cells were rinsed twice with ice-cold PBS (5 min each) and permeabilized with 0.1% Triton X-100 (Sigma-Aldrich) for 20 min at room temperature. After permeabilization, nonspecific binding sites were blocked by incubating the samples in 1% (w/v) bovine serum albumin (BSA) (Sigma-Aldrich) for 45 min at 4 °C. For actin staining, phalloidin (Cell Signaling Technology) was diluted 1:50 in a 0.1% (w/v) BSA solution, and samples were incubated with this solution for 45 min at room temperature in the dark. Excess stains were removed by washing with PBS. Nuclear staining was performed using DAPI (1:1000 in PBS, Biotium), with samples incubated for 5 min at room temperature in the dark. The unbound dye was removed by washing the samples with PBS before imaging. Fluorescence imaging was conducted using a Zeiss LSM 880 inverted confocal microscope, and quantitative analysis of DAPI-stained nuclei was performed with ImageJ software (NIH, Bethesda, MD, USA) to deduce the cell count (cells mm⁻²).

Metabolic activity of i-HNSCs on BIPORES fibers was quantified on days 1, 7, 14, and 21 using a resazurin-based PrestoBlue assay (Invitrogen) according to the manufacturer's instructions. At each time point, the i-HNSC growth medium was aspirated and replaced with a solution containing a 1:10 dilution of PrestoBlue reagent in fresh growth medium. Samples were then incubated for 4 h at 37 °C in a 5% CO₂ humidified incubator to allow viable cells to reduce resazurin to the fluorescent resorufin metabolite. Following incubation, fluorescence measurements were taken with an excitation wavelength of 560 nm and an emission wavelength of 610 nm using a calibrated plate reader. Each condition and time point was assessed in four biological replicates to ensure statistical validity.

Immunostaining: i-HNSCs cultured on BIPORES fibers were processed for immunostaining to detect neuronal and astrocyte differentiation markers. Cells were initially washed twice with 1× PBS and fixed in 4% (v/v) PFA (Thermo Scientific) for 15 min at room temperature. Following fixation, samples were washed three times with ice-cold 1× PBS and permeabilized in 0.3% Triton X-100 (Sigma-Aldrich) in PBS for 15 min at

room temperature. Cells were then washed twice with 1× PBS (5 min each) and blocked in a solution of 1% BSA and 0.1% Triton X-100 in PBS for 1 h at room temperature to minimize nonspecific binding. For primary antibody incubation, a solution was prepared in blocking buffer containing either mouse anti-GFAP (Invitrogen, 14-9892-82) or mouse β-III tubulin (Invitrogen, MA1-118), each diluted 1:200. The samples were incubated in the antibody solution overnight at 4 °C in a humidified chamber. After incubation, cells were washed three times with 1× PBS (5 min each) to remove unbound primary antibodies. Secondary antibody labeling was performed by incubating cells with goat anti-Mouse IgG (H+L) Alexa Fluor 594 (Invitrogen, A-11032) diluted 1:200 in blocking buffer for 2 h at room temperature in the dark. After incubation, cells were washed three times with 1× PBS (5 min each) in the dark to remove excess secondary antibody. Nuclear counterstaining was achieved by incubating cells with DAPI (1:1000 in PBS) for 5 min at room temperature in the dark, followed by a final PBS wash. The stained samples were then visualized using a Zeiss LSM 880 inverted confocal microscope.

Calcium Imaging: Calcium dynamics in differentiated i-HNSCs on BIPORES fibers were assessed on Day 28 using the fluorescent calcium indicator Fluo-8 AM (AAT Bioquest). A stock solution was prepared by dissolving 50 μg of lyophilized Fluo-8 AM in 10 μL of sterile dimethyl sulfoxide (Sigma-Aldrich), yielding a concentration of 5 mM. For staining, this stock was diluted to prepare a working solution containing 5 μM Fluo-8 AM and 0.04% (v/v) Pluronic F-127, the latter facilitating enhanced dye dispersion and cellular uptake. Prior to imaging, i-HNSC differentiation medium was aspirated from neural cultures on BIPORES fibers, and the Fluo-8 AM working solution was added. Samples were incubated at 37 °C in a humidified 5% CO₂ incubator for 30 min to allow for dye uptake and de-esterification. After incubation, the dye solution was replaced with HBBS buffer (Gibco) for imaging. Calcium imaging was conducted using a Zeiss LSM 880 inverted confocal microscope equipped with an FITC filter set (excitation: 490 nm, emission: 525 nm, cutoff: 515 nm). Time-lapse recordings captured dynamic calcium transients over 120 cycles with 1.26 s intervals, yielding a total recording duration of ≈150 s. Data were exported in CZI, TIFF, and MOV formats for analysis and visualization. ImageJ was utilized for postacquisition analysis, and regions of interest corresponding to neurons exhibiting calcium transients were selected visually. Baseline fluorescence (F_n) was determined by averaging intensities during quiescent periods. Fluorescence changes ($\Delta F/F_n$) were quantified using

$$\Delta F/F_0 = \frac{F_t - F_0}{F_0} \quad (2)$$

Fabrication of 3D PEGDA-BIPORES Constructs: Prior to construct fabrication, the microfluidic device was pretreated by coating with a 5% PDAD-MAC solution in 1 M NaCl for 2 h, followed by a resting period of 5 h to ensure stable deposition of the coating. The device was then thoroughly rinsed with 60 mL of deionized water to achieve a uniform surface coating. The coated microfluidic device was mounted on an Allevi-2 3D bioprinter. PEGDA-BIPORES fibers were synthesized using the STRIPS method, as described previously, integrated with the bioprinter setup. Once a stable flow of fiber extrusion was achieved, a preprogrammed G-code path was uploaded to the printer. The bioprinter executed the path for layer-by-layer fiber deposition, repeating the process until the target construct height was reached. Upon completion, water was carefully aspirated from the extrusion bath. The resulting construct was then crosslinked by exposure to 405 nm light using a NEXTTORCH flashlight system at an intensity of 5.5 mW cm⁻² for 30 s, ensuring complete polymerization and stabilization of the 3D architecture. The crosslinked PEGDA-BIPORES constructs were retrieved and processed for subsequent cell seeding or material characterization.

BIPORES Construct Processing for i-HNSC Culture: Following fabrication, PEGDABIPORES constructs underwent static or dynamic postprocessing to remove residual CTAB and establish a conducive environment for iHNSC culture. For static processing, constructs were immersed in 200 proof ethanol at room temperature for three days to extract residual CTAB. The dynamic protocol involved placing constructs in a 37 °C bath containing 200 proof ethanol with continuous stirring for 3 days, enhancing mass

transfer for more efficient CTAB removal. A dual sterilization approach was subsequently employed to prepare the constructs for cell culture. They were transferred to 70% ethanol and exposed to UV light for two days to ensure sterilization. After UV treatment, the scaffolds were thoroughly rinsed with sterile PBS to remove residual ethanol and then equilibrated in growth medium for 24 h prior to iHNSC seeding.

Residual CTAB Quantification: Residual CTAB in processed BIPORES constructs was quantified by UV-vis spectrophotometry. Samples ($n = 3$) were weighed to 20 mg and incubated in 1 mL PBS (pH 7.4) at room temperature for 1 h to extract residual surfactant. Supernatant absorbance at 215 nm was measured using a UV-vis spectrophotometer (Thermo Fisher Scientific) with PBS as blank. CTAB concentration was determined from standard curves ($0\text{--}1000\text{ }\mu\text{g mL}^{-1}$) and converted to mM using CTAB's molecular weight (364.45 g mol^{-1}). Concentrated samples were diluted to maintain absorbance ≤ 2 AU, and the dilution factor was incorporated into the final calculations to report the original sample concentration.

Cytocompatibility Assessments for BIPORES Constructs and Biphasic Model: PEGDA-BIPORES constructs were cut to dimensions of $1500\text{ }\mu\text{m} \times 2000\text{ }\mu\text{m} \times 1500\text{ }\mu\text{m}$. Constructs were placed in individual wells of a 96-well plate to confine the culture environment and enhance cell adhesion. Cells were allowed to adhere for 20 min at $37\text{ }^{\circ}\text{C}$ in a 5% CO_2 incubator before slowly adding i-HNSC growth medium to each well. For hydrogel-encapsulation experiments, 5×10^5 i-HNSCs were seeded per construct. On day 5, cell-laden scaffolds were transferred to fresh wells and embedded either in type I rat-tail collagen (Advanced Biomatrix) or in PEGDA (M_n 700, Sigma). Collagen was prepared at 4 mg mL^{-1} with the supplier's neutralizing solution, fully encasing each construct, and polymerized for 30 min at $37\text{ }^{\circ}\text{C}/5\%\text{ CO}_2$ before medium addition. For PEGDA encapsulation, a 4 wt% PEGDA solution containing 0.25 wt% LAP was added and photopolymerized with two 15 s exposures to 405 nm light, after which fresh medium was supplied. On Day 7, actin/DAPI staining was conducted as previously described to assess cell viability and morphology. Confocal microscopy was performed to acquire Z-stacks through the constructs. 2D projections from these Z-stacks were generated using ImageJ to visualize i-HNSC growth within the engineered matrix.

Statistical Analysis: All statistical analyses and data visualization were conducted using GraphPad Prism version 10 (Boston, MA, USA). Data are presented as mean $\pm$ standard error of the mean (SEM) or standard deviation (SD) from a minimum of four biological replicates. For comparisons between two groups, an unpaired two-tailed t -test was employed. A two-way ANOVA followed by Tukey's multiple comparisons test was utilized to evaluate differences among multiple groups with two independent variables. Statistical significance was defined as $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, and $****p < 0.0001$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

P.D.O. contributed to conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, software development,

validation, visualization, and writing the original draft, review, and editing. K.D. contributed to data curation, investigation, methodology, software support, and writing, review, and editing. S.B.I. conducted the investigation, methodology, and writing, review, and editing. B.D.C. performed formal analysis, investigation, and methodology. A.B. contributed to writing, review, and editing. M.B. contributed to data curation, investigation, and methodology. M.F.H. assisted in writing, review, and editing. B.A. assisted in writing, review, and editing. I.N. contributed to conceptualization, funding acquisition, methodology, project administration, resources collection, supervision, validation, and writing, review, and editing.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

3D neural tissue engineering, bicontinuous scaffolds, bioactive materials, hierarchical microarchitectures, in vitro tissue models, neural stem cells

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